Interpreting Random Survival Forest Using SurvSHAP(*t*) for Patients with Complementary Therapy at Nur Hidayah Hospital, Bantul, Yogyakarta, Indonesia

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Abstract. Machine learning models are highly flexible in handling data complex and improving prediction accuracy, making them a valuable tool for dynamic, time-sensitive cases. However, their lack of interpretability, often termed as "black-box" behavior, poses challenges especially in healthcare, where understanding the reasoning behind prediction is essential. In survival analysis, the Random Survival Forest (RSF) is a popular machine learning model that leverages ensemble learning for accurate predictions but is less interpretable than semi-parametric models such as Cox Proportional Hazard (PH) regression. SuvrSHAP(*t*) address this issue by contribution to the survival function. This study applies survSHAP(*t*) to mental conditions data from Nur Hidayah Bantul Hospital, Yogyakarta, Indonesia, to examine the impact of complementary therapies on patient mental conditions. The research aims to enhance the interpretability of machine learning models in survival analysis, particularly in mental conditions context. The RSF model performed well, achieving C-index scores 0.86 on training and 0.88 on testing data. SurvSHAP(*t*) revealed that age, history of supranatural interactions, loss of consciousness upon hospital arrival, patient reaction when complementary therapy is carried out and the patient's main complaint regarding mental conditions were the most significant predictors of recovery. Interestingly, the impact of these variable shifted over time. For example, patients who arrived unconscious tended to recover faster, while passive reactions to therapy showed varying effects depending on the time point. By understanding these temporal patterns, healthcare providers can design more effective and targeted interventions to improve mental conditions outcomes.

Keywords: RSF, SurvSHAP(*t*), Mental Conditions, Complementary Theraphy.

1 Introduction

The flexibility of machine learning (ML) in handling diverse data types without strict assumptions has made it a popular tool across various fields, including

healthcare[1]. It is only in recent years that there has been a push towards the recognition of the need to have healthcare solutions powered by machine learning. This has led researchers to suggest that it is only a matter of time before machine learning will be ubiquitous in healthcare[2]. However, as machine learning models grow in complexity, the trade-off between predictive accuracy and interpretability becomes more pronounced. This is particularly critical in survival analysis, where understanding the factors influencing time-to-event outcomes, such as patient recovery or mortality, is essential [3]. By explaining the reasoning behind predictions, interpretable machine learning systems give users reasons to accept or reject predictions and recommendations [4].

Random Survival Forest (RSF), an extension of Random Forest[5], has gained popularity in survival analysis due to its ability to handle censored data and provide robust predictions without relying on the Proportional Hazards assumption such as Cox PH regression, offer more flexibility in modeling non-linear patterns and complex interactions[5]. Despite its strengths, RSF is often considered a "black-box" model, ML models can be less interpretable due to their complexity and the intricate relationships between numerous parameters, making it difficult to understand and explain how predictions are derived, which is an important consideration in clinical applications[6]. To address this issue, interpretable machine learning methods based on Shapley theory have been developed. Shapley theory has evolved into SHAP (SHapley Additive exPlanations)[7], and for survival analysis, it has been adapted into SurvSHAP[8]. While SurvSHAP is one of the most effective methods for globally explaining machine learning models, it has limitations when applied to time-dependent explanations. To overcome this, SurvSHAP(t) was proposed as an extension of SHAP specifically designed for survival analysis. This method provides time-dependent explanations of predictions made by models such as Random Survival Forests (RSF), offering deeper insights into the temporal dynamics of survival outcomes [9].

Mental health disorders, which affect one in eight people globally, have seen a significant rise due to the COVID-19 pandemic, with anxiety and depression increasing by 26% and 28%, respectively, in 2020 [10]. In Indonesia, over 19 million people suffer from emotional mental disorders, highlighting the urgent need for effective interventions [11]. Complementary therapies, such as Quranic therapy, are increasingly used to support mental conditions treatment. At Nur Hidayah Hospital in Yogyakarta, Indonesia, Quranic therapy is employed to aid patients in managing mental conditions. However, evaluating the effectiveness of such therapies requires advanced analytical tools. Survival analysis, particularly when combined with interpretable machine learning methods like RSF and SurvSHAP(t), can provide insights into the factors influencing recovery and help optimize treatment strategies.

This study applies RSF and SurvSHAP(t) to analyze mental conditions data from patients receiving Quranic therapy at Nur Hidayah Hospital. The research aims to (1) predict recovery time using RSF and (2) enhance the interpretability model's predictions using SurvSHAP(t) to identify key factors influencing recovery. By bridging the gap between accuracy and interpretability, this study contributes to the development of complementary therapies and advances the application of machine learning in survival analysis. The findings are expected to provide actionable insights for mental conditions treatment and enhance understanding of interpretable machine learning in healthcare.

2 Literature Reviews

Previous research by Arlianni (2023) identified Stratified Cox regression as the best model for predicting the repair time of mental condition in patients undergoing complementary therapy at Nur Hidayah Hospital. This conclusion was based on the model's smallest Akaike Information Criterion (AIC) value of 286.839, indicating a better fit compared to other approaches especially Cox PH regression. Additionally, studies by Jabbari (2020) and Diego-Cordero (2022) demonstrated that spiritual interventions, such as Quranic recitation, positively impact mental health. However, Aljawadi (2020) found no significant differences in mental health outcomes based on gender.

Despite its effectiveness, Cox Proportional Hazard (PH) regression has limitations, particularly its assumption of a constant hazard ratio over time, which may not always hold in real-world data. To overcome this, Random Survival Forest (RSF) has been introduced as an alternative that offers greater predictive performance. Studies by Rahman et al. (2023) and Zhao et al. (2024) showed that RSF achieved a high C-index of 85.5% in predicting mortality based on physiological factors. Additionally, Suantari (2023) compared RSF with SSVM-RBF and SSVM-ADD, concluding that RSF is superior in handling mixed variable types.

However, a key drawback of RSF is its lack of interpretability, as it does not directly indicate the most influential factors in its predictions. Previous research conducted by Sarica et al., (2023) interpreted RSF with SHAP (SHapley Additive exPlanation) in cases of Alzheimer's disease, but this method is less suitable for survival analysis because it does not adequately account for censored data. To improve interpretability, SurvSHAP was developed by Alabdallah et al. (2022), offering a more effective explanation framework for non-linear survival models compared to linear models like Cox PH.

Building on this, SurvSHAP(t) was introduced by Krzyziński et al. (2023) to capture temporal effects in survival models. This method allows for the analysis of variable contributions to survival probabilities across different time points. SurvSHAP(t) was validated on simulated survival distributions and heart failure cases, proving to be more stable than SurvLIME.

In this study, SurvSHAP(t) is applied to mental conditions patients undergoing complementary therapy based on Quranic recitation. By integrating RSF with SurvSHAP(t), this research aims to enhance the interpretability of survival analysis models. To the best of our knowledge, this is the first study to apply SurvSHAP(t) in the context of mental conditions patients receiving complementary therapy, providing novel insights into the key factors influencing survival outcomes in this population.

3 Research Methods

The research methodology includes comprehensive descriptions of the study framework, including data source, variable descriptions, and data analysis techniques.

3.1 Data Source

The data on the number of patients with mental disorders who received Al-Quran therapy treatment in this study was divided into two parts, namely training and testing. Training data is used to build the model. Test data is used to determine the approximate accuracy of the model built from the training data.

Table 1. Table captions should be placed above the tables

Notation	Variable Name	Description
T	Length of time to improve Patient's Mental Conditions	Days
δ	Status	0: Censored; 1: Recovery (event)
X_1	Patient Condition	0: Normal; 1: anxious; 2: loss of consciousness
X_2	Disruption of worship	0: No distractions 1: Lazy Worship 2: No Worship
X_3	Supernatural interaction	0: None; 1: Never had; 2: Has interaction
X_4	Gender	0: Male; 1: Female
X_5	Age	(Years)
X_6	Marital Status	0: Single; 1: Married
X_7	Education	0: Elementary School; 1: Junior High School; 2: Senior High School; 3: College
X_8	Main complain	0: Mental; 1: Physical
X_9	History of chronic disease	0: None; 1: Ever had
X_{10}	Therapy reaction	0: Passive; 1: Active

One important key in survival analysis is to pay attention to the event interest and the time of observing patients from the beginning of therapy until the occurrence of the event (recovery). Sometimes not all patients can be observed until the end of the time observation, which results in censored data. To determine whether the observation is to be censored or not is through the following definition.

The observed event is the improvement of the patient's mental condition after complementary therapy, indicated by a decrease in all SGR (X_1 Condition Patient, X_2 Disruption of worship, X_3 Supernatural Interaction) which values to zero, which is considered normal.

1. Patient Condition (S) is the first SGR score. This predictor variable is an assessment of the patient's condition upon arrival in hospital for complementary therapy.
2. Worship Disruption (G), is the second SGR score. This variable assesses the disruption of worship, specifically the obligatory prayers experienced by patients upon their arrival for complementary therapy.

3. Supernatural interaction (R) values from the category of supernatural interaction history that explain the history of interaction with patients and families.

   Here are some combinations as an illustration of determining data experiencing events or being censored as presented in the Table 2.

Table 2. Determination of Event

Patients	SGR Scores Before Treatment			SGR Scores After Treatment			Data Status
	S_b	G_b	R_b	S_a	G_a	R_a	
A	1	2	0	0	0	0	Event
B	1	1	0	0	1	0	Censor
C	2	2	2	1	0	1	Censor
D	1	2	3	0	0	0	Event
E	0	1	0	0	0	0	Event
F	1	2	2	1	0	0	Censor

3.2 Data Processing

To achieve the objectives of the study, the analysis process steps are illustrated in Figure 1.

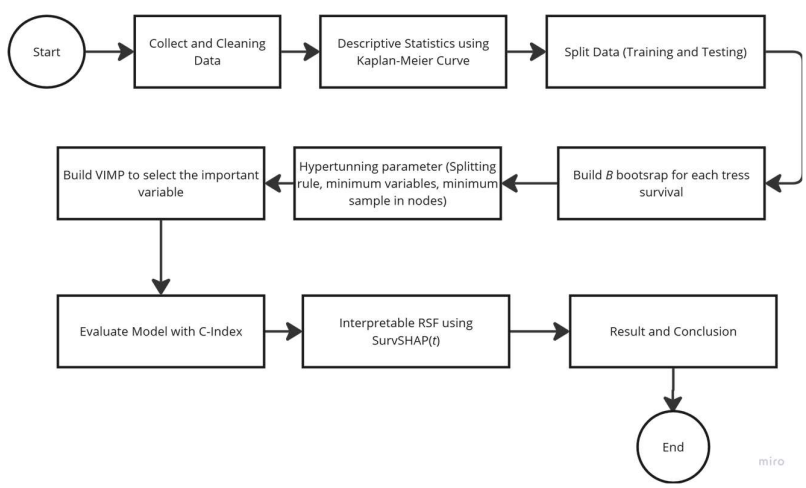


Fig. 1. Workflow of the proposed survival analysis framework. The framework includes data preprocessing, Kaplan–Meier analysis, model development using Random Survival Forest, hyperparameter tuning, variable importance selection, performance evaluation with the C-index, and model interpretation using SurvSHAP(t).

The process begins with the collection and cleaning of data, ensuring that the dataset is accurate and suitable for analysis. Once the data is prepared, the next step is splitting data into training and testing. Following this, determine parameters for the Random Survival Forest (RSF) and build model from data training. These parameters include the splitting rule, which dictates how the data is divided at each node, the minimum node size required to split a node, and the minimum sample size needed at a leaf node. After configuring the model, it is evaluated to assess its predictive performance. This evaluation helps in understanding how well the model can forecast outcomes based on the given data. Finally, the model's predictions are explained to gain insights into the factors that influence the outcomes. This step may involve using techniques like survSHAP(*t*) (Survival SHapley Additive exPlanations) to interpret the model's decisions and understand the contribution of each feature to the predictions. This structured approach ensures a comprehensive development and understanding of the Random Survival Forest model, from data preparation to model interpretation.

Random Survival Forest (RSF). Survival analysis is a statistical method used to analyze time-to-event data [12], focusing on tasks that involve censored data, where incomplete information about an individual's survival time is present. Censoring schemes are divided into three types: right censoring, left censoring, and interval censoring. In this study, right censoring is used, which occurs when a subject withdraws, does not experience the event of interest by the end of the study, or is lost to follow-up. Mathematically, a given instance i is represented as a triplet $(\mathbf{x}_i, y_i, \delta_i)$, where $\mathbf{x}_i = [\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_p]$is the variable vector, δ_i is the indicator of event, when δ_i=1 is survival time T_i and δ_i=0 is censoring time C_i and y_i is the observed time. The first key object is the survival function (1) which describes the probability of an individual surviving until time *t* without experiencing the event, i.e.,

$$(t) = P(T > t) = 1 - P(T \leq t) = 1 - \int_0^t f(x)dx = 1 - F(t) \tag{1}$$

In addition to the survival function, the hazard function is also the basis of survival analysis where this function describes the risk of an event occurring at a certain time with the assumption that the observed individual has survived until that time.

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t | t \geq T)}{\Delta t} \tag{2}$$

The relationship between survival function and hazard function can be formulated in equation 3 below.

$$S(t) = e^{-H(t)} \tag{3}$$

The Kaplan-Meier estimator, also known as the product limit estimator, is a non-parametric statistic used to estimate the survival function from time-to-event data. It provides a way to handle censored data, which is a common feature of survival analysis[13]. Mathematically, the KM form can be written as follows.

$$\hat{S}(t_{f-1}) = \prod_{i=1}^{f-1} \hat{P}_r(T > t_i | T \geq t_i) \tag{4}$$

where $\hat{S}(t_{f-1})$ is the probability of survival past the previous failure time and $\prod_{i=1}^{f-1} \hat{P}_r(T > t_i | T \geq t_i)$ is the probability of subjects surviving past time t_i under the condition that they have survived until time t_i.

Machine Learning in survival analysis such as Random Survival Forest (RSF) adapts the Random Forest (RF) method so that it can handle censored data and estimate survival time. This is what distinguishes it from other supervised learning methods, such as classification and regression. The fundamental difference between Random Forest and Random Survival Forest (RSF) lies in the ability of RSF to consider censored data and estimate survival time. The Random Survival Forest algorithm can be explained through the following flowchart.

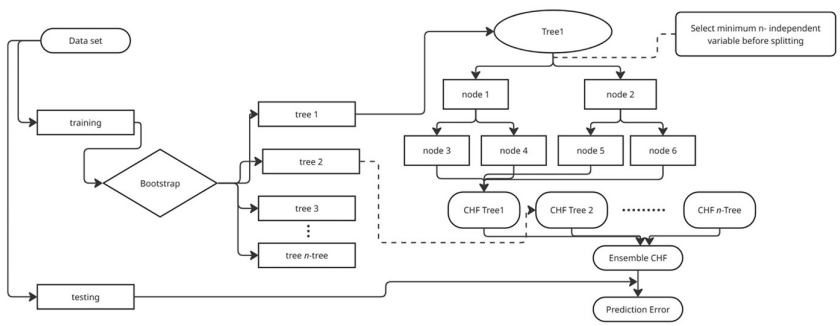


Fig. 2. Random Survival Forest (RSF) algorithm workflow. The workflow illustrates data splitting, bootstrap-based tree construction, random variable selection at each node, ensemble cumulative hazard estimation, and prediction error evaluation.

Similar to the Random Forest method, survival trees are built based on the same principles. This method is an advanced method of the random forest method introduced by Breiman (2001). There are nodes, leaf nodes and as many iterations as possible with random elements when selecting random variables. Figure 2 explains the process of how the Random Survival Forest flow can be organized in the following algorithm.

1. The dataset is explicitly separated into training and testing (70% and 30%).
2. Draw B bootstrap samples from the training data set.
3. Build Survival Trees with the following important notes:
   - Each tree is built using bootstrap samples from the training set.
   - At each node, randomly select $\sqrt{p}$ from p covariate
   - Using a split rule such as log rank to maximize the difference between child nodes.
   - Continue building the tree until it meets the constraint that the terminal node must have at least $d_0 > 0$ unique deaths.
4. Calculate the Cumulative Hazard (CH) for each tree based on the training data.
5. Calculate the average CH of all trees to get the ensemble prediction.
6. Use testing data for C-index validation.

However, the fundamental difference between RF and RSF is in the separation of nodes when building a tree. In RSF, one of the rules used as a measure of impurity is the log rank statistic.

Log Rank Splitting Rule. Suppose a node h can be split into two daughter nodes α and β. The best split at a node h, on a covariate x at a split point c^* is the one that gives the largest log-rank statistic between the two daughter nodes [14]. The log-rank statistic for a split on x at a given covariate value c^* is defined as equation 5.

$$i(x, c^*) = \frac{\sum_{j=t_1}^{tN}(d_{\alpha,j} - E(D_{\alpha,j}))}{\sqrt{\sum_{j=t_1}^{tN} Var(D_{\alpha,j})}} \tag{5}$$

where $d_{\alpha,j}$ is the number of events in daughter node α at time point j. The expected number of events in daughter node α, $E(D_{\alpha,j})$ and its variance are given by

$$E(D_{\alpha,j}) = R_{\alpha,j}\frac{d_j}{R_j} \tag{6}$$

$$Var(D_{\alpha,j}) = \frac{R_{\alpha,j}}{R_j}\left(1 - \frac{R_{\alpha,j}}{R_j}\right)\left(\frac{R_j - d_i}{R_j - 1}\right)d_j \tag{7}$$

where d_j is the total number of observed events at time point j. $R_{\alpha,j}$ is the number of individuals at risk in node α at time point j and R_j the combined number at risk in daughter nodes α and β[15].

Variable Importance. The permutation feature importance measurement, also known as variable importance (VIMP), was introduced by Breiman (2001) in the context of Random Forest. The idea behind VIMP is that for a variable not associated with the response, permuting its values will not affect the prediction error, so the estimated VIMP will be close to zero[16].

$$VIMP = \frac{1}{N_{\text{tree}}}\sum_{t=1}^{Ntree} Error_{OOB,t}(\mathbf{x}_j) - Error_{OOB,t} \tag{8}$$

Where the $Error_{OOB,t}$is Error prediction in Out-of-Bag (OOB) data for tree *t*-th and without randomization and $Error_{OOB,t}(\mathbf{x}_j)$ is error prediction in OOB after $(\mathbf{x}_j)$is randomized.

C-Index. The C-Index has been extended to the survival analysis setting, with Harrell or Uno's C-Index definitions being common choices for the validation of survival models[17]. The C-Index is equal to the prediction error rate, where the prediction error rate is expressed by equation 9.

$$1 - (C\ Index), 0 \leq E \leq 1 \tag{9}$$

where E is the error rate. If $C\ Index$ is closer to or equal to 1, it can be interpreted as $error\ rate$=0, indicating perfect prediction[18]. Mathematically, C-index can be written in equation 10 as follow.

$$C\ index\ = \frac{\sum_{i<j}(1_{T_i<T_j} \cdot 1_{\eta_i>\eta_j})\delta_i + (1_{T_i>T_j} \cdot 1_{\eta_i<\eta_j})\delta_j}{\sum_{i<j}(1_{T_i<T_j})\delta_i + (1_{T_i>T_j})\delta_j} \tag{10}$$

For each distinct patient i and j (with $i \neq j$), the risk score (η) will be compared to the time of event occurrence (T). If T_i and T_j are uncensored, then we can see when both patients have the disease. A pair (i, j) is called a concordant pair (CP) if $\eta_i > \eta_j$ and $T_i < T_j$ so the value is 1, otherwise it is 0, and a discordant pair (DP) if $\eta_i > \eta_j$ and $T_i > T_j$. However, if T_i and T_j are censored, then we do not know who got the disease first. Therefore, we do not include this pair in the calculation.

Interpretable Machine Learning. Interpretability refers to the degree to which a human can understand the cause of a decisions[19]. Another definition is the degree to which a human can consistently predict the model's outcome[20]. One of the interpretable models is model-agnostic methods, which are suitable for survival analysis, such as SurvSHAP(t). The model generalizes SHapley Additive exPlanations (SHAP)[7] to survival models. SurvSHAP(t) uses the concept of Shapley values from game theory to calculate the contribution of each variable to the model's predictions. These Shapley values are calculated for each point in time, resulting in a function that shows how each variable affects predictions over time. The advantage of this method is that SurvSHAP(t) is able to detect variables that have time-dependent effects, which previous explanation methods, such as SurvLIME[21].

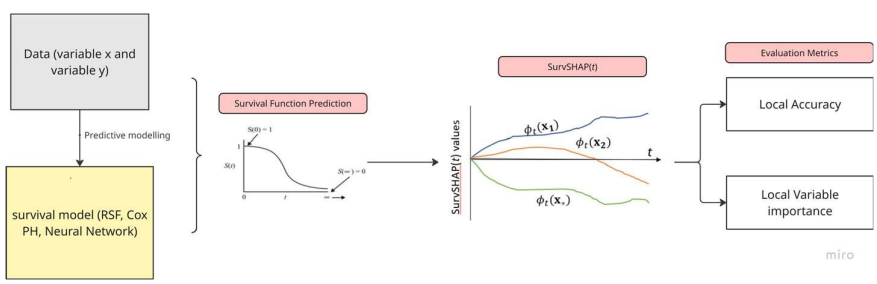


Fig. 3. The workflow of SurvSHAP(t) which requires a survival dataset and a method such as RSF, Cox Model or Survival Neural Network.

Figure 3 shows that SurvSHAP(t) allows for time-dependent explainability of any survival model predictions. SurvSHAP(t) values add up to the survival function predicted by the model and aggregated over time can be treated as a local importance ranking of variables[9].

For each observation $\mathbf{x}_*$, SurvSHAP(t) calculates the contribution of the variable to the prediction of the survival function using the following formula.

$$\phi_t(\mathbf{x}_*, d)\ = \frac{1}{|\Pi|}\sum_{\pi\epsilon\Pi}\left(e_{t,\mathbf{x}_*}^{before\ (\pi,d)\cup\{d\}} - e_{t,\mathbf{x}_*}^{before\ (\pi,d)}\right) \tag{11}$$

where Π is the set of all permutations of p variables, π is one of the permutations, before (π, d) is the subset of variables that appear before variable d in permutation π

and $e^{D}_{t,\mathbf{x}_*}$ is the expected value of the survival function $\hat{S}(t)$ under the condition that the variables in subset D have the same value as $\mathbf{x}_*$. To facilitate comparison between different models and time points, SurvSHAP(t) values can be normalized to a common scale (from -1 to 1) using the formula.

$$\phi^*_t(\mathbf{x}_*, d) = \frac{\phi_t(\mathbf{x}_*, d)}{\sum_{j=1}^{p} \phi_t(\mathbf{x}_*, j)} \tag{12}$$

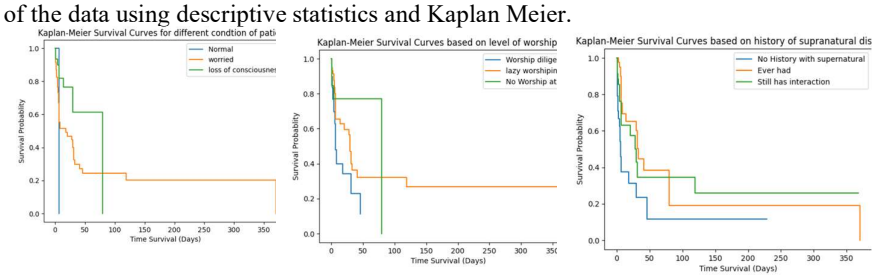
SHAP value for each variable that is close to 1 tends to make a positive contribution to prediction, while if the value is closer to -1 then the variable tends to make a negative contribution to survival prediction. $\phi^*_t(\mathbf{x}_*, d)$ indicates the normalized SurvSHAP(t) value for variable d at time t. $\phi_t(\mathbf{x}_*, j)$ is he SurvSHAP(t) value for variable j at time t. To speed up the calculation, SurvSHAP(t) can also be estimated using Shapley kernel and weighted linear regression with functional response and scalar variables. The estimation formula is written in equation 13.

$$\Phi = (Z^T W Z)^{-1} Z^T W Y \tag{13}$$

where Z is is a matrix of all binary vectors representing the coalition of variables. W is a diagonal matrix containing the weights of the Shapley kernel. Y is a matrix containing the survival function values predicted by the F model for each row of the Z matrix. Φ represents a results matrix containing explanations for each variable in the model.

4 Results and Discussion

Mental Conditions Recovery Predictions Analysis Results. Before the Random Survival Forest (RSF) analysis is carried out, it is necessary to observe at the characteristics of the data using descriptive statistics and Kaplan Meier.



(a) (b) (c)

Fig. 4. Kaplan-Meier Survival Curves based on patient condition. Panel (a) shows survival probabilities by patient condition, panel (b) by level of worship, and panel (c) by history of supernatural disturbances. Differences in survival patterns across groups are observed over the follow-up period.

Picture 4 is a Kaplan-Meier Plot to see the important characteristics of patients with mental disorders who received complementary Al-Quran therapy. Plot 4 (a) illustrates

the Kaplan-Meier (KM) of variable X_1, the condition of the patient when coming to the hospital when first diagnosed with a mental disorder. The three survival curves in the Figure intersect each other, indicating that patients with different attitudinal conditions have different chances of experiencing improvement in their mental state. Plot 4 (b) illustrates the Kaplan-Meier curve of variable X_2 which is the level of worship craft of the patient receiving therapy. This variable is obtained when the patient will undergo complementary therapy, the value of worship disorder measures the condition of the patient's worship, meaning whether the worship of the complementary therapy patient has a disorder. Plot (c) shows the KM curve of variable X_3. This variable measures the patient's history of interaction with the supernatural, whether the complementary therapy patient has no supernatural interaction, has had or still has supernatural interaction. The three curves in Plot c intersect, indicating that patients with different interaction histories have different probabilities of experiencing improvement in mental state.

The average age of patients undergoing complementary therapy at Nur Hidayah Hospital is 40.86 years, with an age range between 2 and 79 years. In this study, based on observed events, 54 patients showed an improvement in mental condition (improved event), while 48 patients were censored due to no improvement.

Among patients who arrived at the hospital in an unconscious state, only 27% experienced mental improvement. Meanwhile, among those who regularly performed the five daily obligatory prayers, 67% showed mental improvement, whereas only 31% of those who did not pray experienced improvement.

Patients with a history of interactions with supernatural elements had a mental improvement rate of 43%. Additionally, male patients had a higher percentage of mental improvement compared to female patients, with a rate of 61%.

Regarding marital status, more than 50% of patients experienced mental improvement, with unmarried patients showing a higher improvement rate compared to married patients. In terms of education levels, all categories had an improvement rate above 50%, except for junior high school graduates, who only reached 20%.

Patients with severe complaints, whether mental or physical, had the same improvement rate of 53%. Furthermore, patients without a history of chronic illness had a higher improvement rate of 56%. Finally, patients who responded passively to therapy had a higher rate of mental improvement compared to those who reacted actively, with a percentage of 74%.

Based on the steps established through the methodology, the RSF model can be built with parameter tunning adjustments, including: The number of trees in the forest is 1000, in order to obtain stable estimates and reduce model variability. The minimum number of samples to perform splits at nodes is 6, while the minimum number of samples at leaves is 3 to ensure each leaf node has enough data to perform predictions. The maximum features considered for each split are set to the square root of the total number of features to reduce the risk of overfitting while improving computational efficiency and the splitting rule use log-rank test. Reproducible random values were set with “random_state=42”, ensuring the experimental results are repeatable. The prediction results for each patient were selected 5 patients to be depicted on the survival curve. These survival curves provide a visual representation of how the probability of survival varies between individuals and over time.

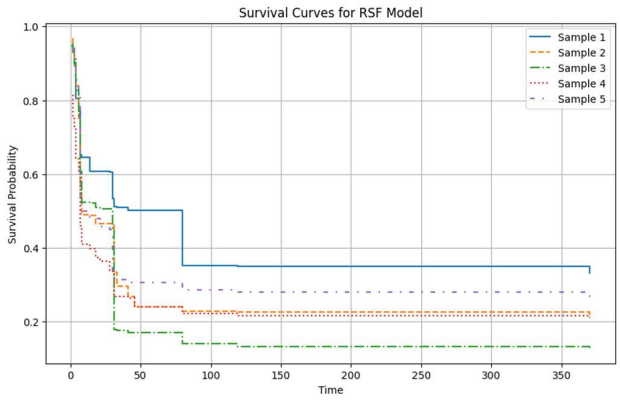


Fig. 5. Plot Sample Survival Curves Prediction based on RSF model. The figure shows individual survival function predictions for five representative patients (Samples 1–5). Each curve represents the estimated time to recovery in mental health under complementary therapy, illustrating patient-specific differences in survival probability over time.

Figure 5 depicts the survival curve generated from the Random Survival Forest (RSF) model. These survival curves depict the survival probability of several individual samples (Sample 1 to Sample 5) as time passes. At the beginning of time (Time = 0), all curves start with a survival probability of 1 (100%), which indicates that at the beginning of the observation, all individuals still survive. Differences in the shape and slope of the survival curves between samples indicate variability in individual risk. This could be due to differences in characteristics or predictor variables that affect the probability of survival. Sample 1, with a steeper decline, may have characteristics that increase risk, while Samples 4 and 5, with a slower decline, may have characteristics that are protective or reduce risk. At times around 50-100, there is a significant drop in survival probability for some samples (especially Sample 1). This may indicate a critical period where the risk of experiencing an event increase.

Feature importance analysis (VIMP) was conducted using Permutation Importance to evaluate the influence of each feature on the performance of the RSF model.

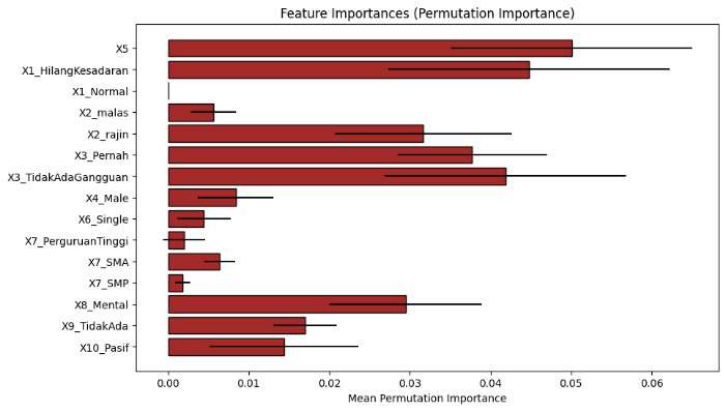


Fig. 6. Permutation variable importance (VIMP) from the Random Survival Forest model. Variables are ranked according to their permutation importance scores, reflecting their relative influence on survival predictions in the Random Survival Forest model.

Figure 6 shows that features X_5 (Age), X_1 (Condition)_Loss of Consciousness, and X_3 (Spiritual)_No supernatural interference have the highest importance values, indicating a significant contribution to the survival prediction model. In contrast, features such as X_1_Normal, X_7 (Education)_SMP (Junior High School), and X_{10}_Passive reaction have low importance values, indicating minimal contribution to the model. In addition, the resulting error bars show variations in the importance estimates, especially for features such as X_1_Lost Consciousness and X_3_No supernatural interference. This reflects the uncertainty that may be caused by the permutation process.

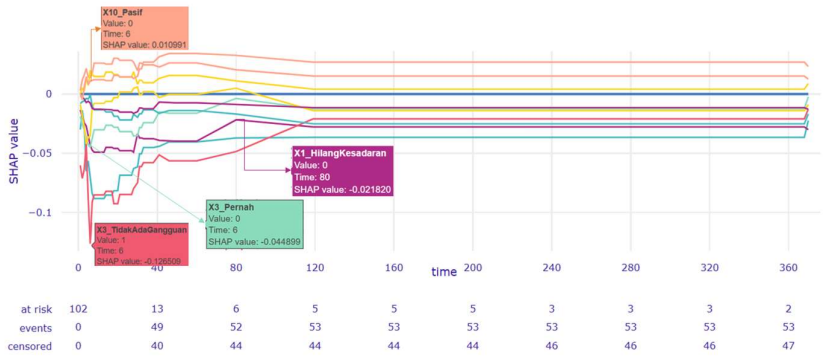
To evaluate the goodness of the RSF model that is formed both on the training data and testing data, the C-index model evaluation is used. Table 3 shows the C-index comparison on the RSF model.

Table 3. Model evaluation using C-Index

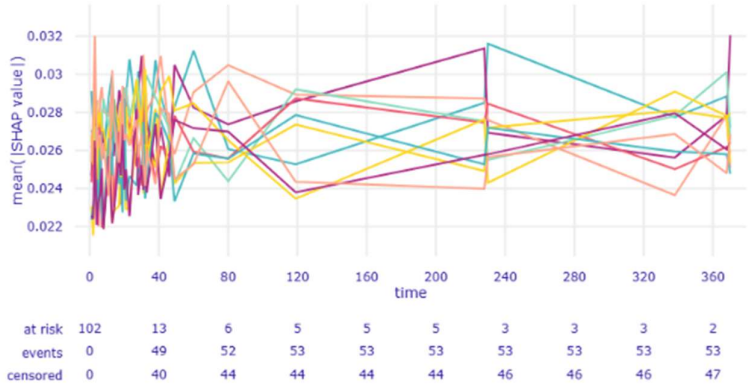
Data	C-index Value
Training	86.61%
Testing	88.21%

On the training data, the C-index value reached 86.61%, indicating the model's ability to predict with high accuracy on the data used for training. Meanwhile, on the test data, the C-index value of 88.21% indicates that the model is able to generalize well to data that has never been seen before. The small difference between these two values indicates that the model does not experience overfitting and has a consistent level of accuracy on both training and testing data.

SurvSHAP(t) Model of Mental Patient Recovery. SurvSHAP(t) analysis was used to evaluate the contribution of features to temporal survival risk prediction.



(a)



(b)

Fig. 7. Time-dependent feature contributions based on SurvSHAP(t). Panel (a) Individual SurvSHAP(t) trajectories illustrating time-varying contributions of selected variables to the survival prediction for a representative patient. Panel (b) Mean SHAP values over time, summarizing the average magnitude of feature contributions across observations.

Plot (a) in Figure 7 contains “at risk” which shows the number of individuals who are still at risk (not cured or censored) at each observation time. The number of patients still under observation (at risk) decreases over time, which is normal in survival/recovery studies. In the initial interval, there were 102 patients at risk, but this number decreased to 2 in the last time interval. This decrease is due to patients who recovered or were censored during the observation period.

“Events” is the number of events that occurred at a given time. In general, it can be seen that the number of patients who experienced recovery (events) increased over time, but then reached a stable point after a certain time interval. This can be seen from the "events" (recovery) column which shows an increase from 0 to 49 in the initial interval, and then stabilizes at 53 after a time interval of 120. This indicates that most

recoveries occur in the initial period of therapy, and after that, the recovery rate tends to stagnate.

The "Censored" label is the number of individuals who were censored (did not recover until the end of the time period). The number of patients who were censored (did not recover within the observation period) also increased over time, but at a slower rate than those who recovered. In the initial interval, there were no censored patients, but the number increased to 47 in the last time interval. This suggests that the longer the observation time, the more patients may drop out of the study before achieving recovery.

In general, the red line curve shows X_3_TidakAdaGangguan (No supernatural interference), the SHAP value is consistently negative at most times, indicating that this feature plays a protective role against the risk of events (extending survival time). On day 6, patients with no supernatural interference had the highest effectiveness with complementary Quran therapy compared to patients with supernatural interference. This can be shown by the SHAP value of -0.1265, which means that patients with a history of not having interactions with supernatural disorders recover faster than patients who are having interactions with supernatural disorders. In comparison, the orange curve (X_{10}_Passive) is a variable that indicates the patient's lack of reaction to the complementary therapy. This curve is always above 0 (positive SHAP) from the start of patient observation until the end of the study. This suggests that patients with a history of passive reactions to complementary therapies have a longer effect on the recovery time of patients with mental disorders.

Plot 7b shows that the average SHAP value in survSHAP(t) is around 0.22 and 0.32 as the study time is observed. This shows relatively small fluctuations over time. This narrow range indicates that the contribution of the predictor variables to the model did not change drastically over the study period and had a consistent influence on the model predictions, so these values are considered relevant and reliable.

5 Conclusion

In this study, age is the most influential predictor of recovery time for mental conditions patients undergoing Quran-based complementary therapy, as identified through Variable Importance (VIMP) analysis. The Random Survival Forest (RSF) model with SurvSHAP(t) interpretation achieved a high predictive performance (C-index: 86-88%), indicating strong reliability in survival time estimation.

Key temporal patterns were observed: The effect of predictor variables on recovery time fluctuated in the early phase (0-120 days) and stabilized afterward. Features with negative SHAP values were consistently associated with shorter recovery times, while positive SHAP values indicated delayed recovery. Notably, patients without supernatural interference showed faster recovery, suggesting that this factor may enhance the effectiveness of complementary therapy. Conversely, patients with passive reactions exhibited prolonged recovery times, underscoring the potential importance of engagement levels in therapeutic outcomes.

These findings have important clinical implications, particularly in personalizing complementary therapy for mental conditions. The ability of SurvSHAP(t) to track

temporal effects offers a dynamic perspective on treatment response, which may assist in tailoring interventions based on individual patient profiles. This insight could be valuable for designing adaptive therapy protocols where treatment strategies evolve over time based on observed patient responses.

Despite these promising results, limitations exist. The log-rank score-based splitting criterion in RSF may not always be optimal; thus, future studies should explore alternative splitting rules. Additionally, the impact of therapy session frequency was not explicitly modeled, warranting further investigation into its role in recovery trajectories. To enhance generalizability, longitudinal survival analysis and multi-center studies should be conducted. Finally, comparing single vs. combination therapies and integrating causal inference methods could provide deeper insights into the effectiveness of Quran-based interventions.

By integrating SurvSHAP(t) for survival interpretability, this study represents a novel application in mental health research and offers a pathway for explainable ML in clinical decision-making. Future research can build upon these findings to refine predictive models and optimize complementary therapy strategies for mental health patients.

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